

GQUENCE

Clinical Nutrition Intelligence for Oncology

Personalised nutritional prescribing, manufacturing and outcome tracking —
in one clinically governed platform.

Pilot evidence · 45 patients · 925 monitoring records

Malnutrition is treated as a side issue

Yet it decides whether treatment can continue

1

Cachexia is near-universal

Up to 80% of advanced cancer patients develop malnutrition; it is implicated in roughly 20% of cancer deaths.

2

Nutrition plans are manual

Calculated by hand, inconsistently, and rarely revised as the patient changes through treatment cycles.

3

Nobody sees deterioration early

Weight loss, falling albumin and rising inflammation are noticed at the next visit — often weeks late.

4

No proof it was delivered

Whether the patient received or consumed the supplement is usually unknown and unrecorded.

The cost of getting it wrong



Treatment delays

Dose reductions and interruptions



Longer admissions

Higher complication rates



Poorer tolerance

Reduced response to therapy



Lost trial data

Nutrition rarely captured rigorously

One platform, three clinical jobs

From prescription through manufacture to measured outcome



PRESCRIBE

A rules-based engine computes an individualised nutritional prescription from anthropometry, biochemistry, cancer type and the active chemotherapy regimen.



MANUFACTURE

The approved prescription becomes a batch-tracked product with two-stage store approval, label printing and full dispatch chain of custody.



TRACK

Weekly reviews recompute the prescription; daily logs record actual intake. Deterioration is flagged automatically against ESPEN / GLIM criteria.

Every step is auditable: who calculated it, who approved it, when it was made, when it reached the patient.

A clinical engine, not a calculator

Every constant is configurable and version-controlled — not hard-coded

136

configurable
clinical parameters

16

chemotherapy & radiation
interaction rules

19

automated safety
alerts

4

formulations for
distinct phenotypes

What the engine reasons over

- Anthropometry — weight, height, BMI, MUAC, grip strength
- Biochemistry — albumin, CRP, haemoglobin, glucose, creatinine
- Disease — cancer site, stage, ECOG performance status
- Therapy — active regimen, radiation field, treatment phase
- Symptoms — dysphagia, mucositis, nausea, GI toxicity

Four clinical formulations

- | | |
|---------------------|--|
| ● Standard | Baseline oncology formulation |
| ● Hydrolyzed | Peptides for GI toxicity / malabsorption |
| ● Diabetic | Reduced carbohydrate, higher fat ratio |
| ● Anti-inflammatory | Elevated omega-3 for high CRP states |

It catches what a busy clinic misses

Drug–nutrient interactions surfaced at the point of prescribing

Worked example — Cisplatin

The engine detects the regimen and raises the associated protocol automatically:



Renal magnesium wasting

Mandatory magnesium protocol initiated



Nephrotoxicity risk

Weekly creatinine monitoring enforced



Escalation trigger

Creatinine > 1.5 mg/dL — hold Cisplatin, escalate to nephrology



Protein adequacy

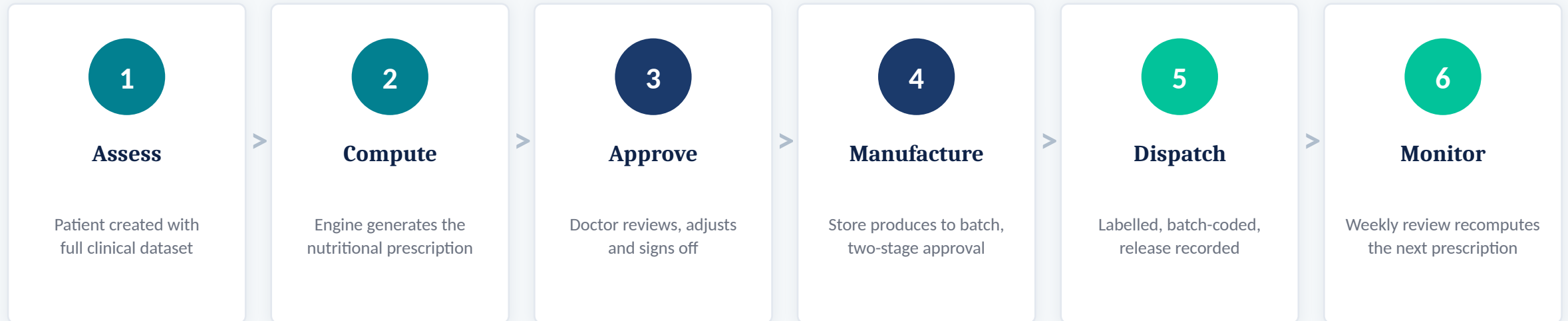
Minimum threshold raised to counteract catabolism

Also covered

- Gemcitabine
- Taxanes
- Fluoropyrimidines
- Irinotecan
- Pemetrexed / antifolates
- Bevacizumab
- Pembrolizumab
- R-CHOP components
- Bortezomib · Lenalidomide
- Olaparib (PARP inhibitor)
- Pelvic / abdominal radiation

A governed chain of custody

Six roles, every transition timestamped and attributed



Step 6 feeds step 2 — the prescription adapts every week rather than being set once at diagnosis.

Roles: Doctor • Assistant • Store Manager • Store Approver • Admin • Super Admin

The prescription changes as the patient does

A weekly review recomputes targets from fresh measurements

What each weekly review captures

- **Weight, BMI, MUAC** anthropometric trajectory
- **Hand grip strength** functional muscle status
- **Albumin, CRP, glucose** nutritional & inflammatory state
- **ECOG** performance status
- **Oral intake %** what the patient managed to eat
- **Supplement compliance %** what was actually consumed

Then it recalculates

Energy & protein targets

Recomputed against current weight and catabolic state

Ingredient recipe

Protein, MCT, omega-3, glutamine quantities regenerated

Formulation switch

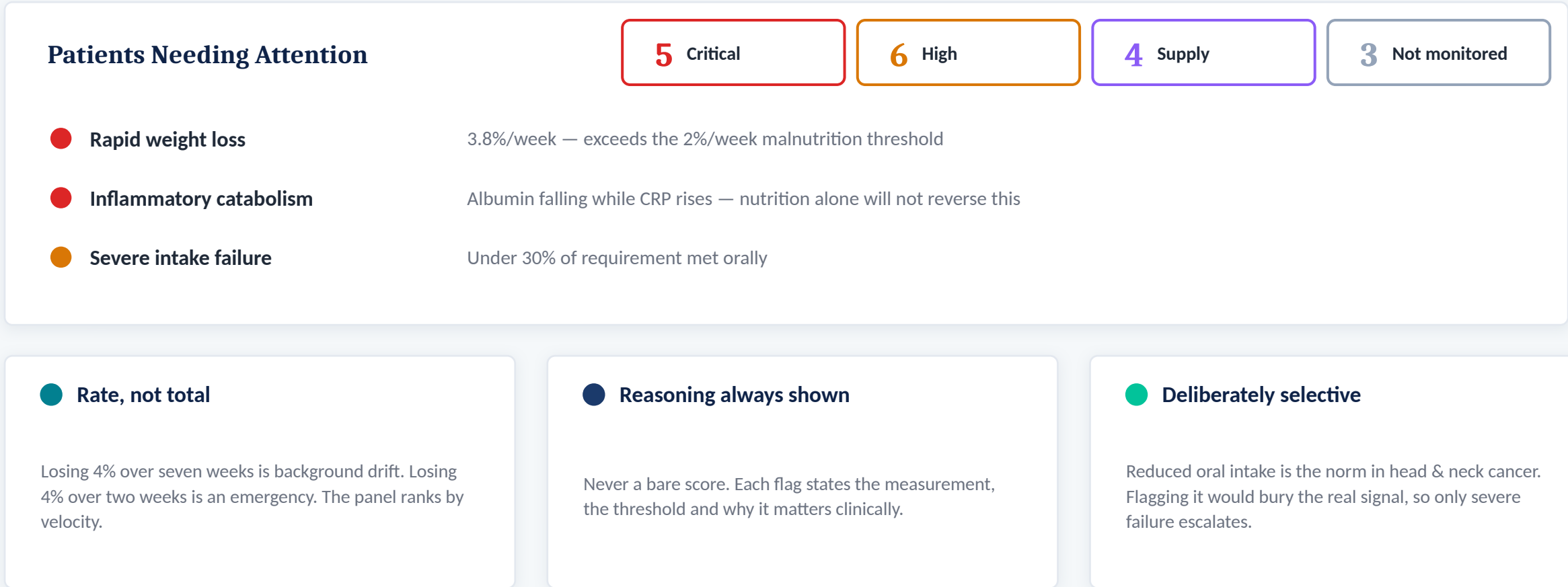
Moves to hydrolyzed or anti-inflammatory as the phenotype changes

New batch code

Week-stamped, traceable to that specific prescription.
Requires doctor approval before it reaches manufacturing.

It tells you who is deteriorating — today

Ranked against ESPEN / GLIM criteria, with the reasoning shown



Proof the patient actually received it

Day-by-day adherence, and the supply gap predicted before it happens

Adherence, measured honestly



31

days on
supplement

93%

adherence of
logged days

2

longest gap
(days)

A day with no log counts as unmonitored — never silently as a missed dose.

Supply predicted, not reported

Each dispatched product covers seven days. The platform knows when every patient runs out — and whether anything is in production to replace it.

2 days left nothing in production

Out of supply 3 days past cover

Covered next week already approved

Treatment gaps get prevented, not documented after the fact.

Research-grade data as a by-product of care

No separate CRF, no retrospective chart review



Enrollment log

Study IDs, consent dates, cohort status derived automatically



Cohort dashboard

Recruitment curve, per-site comparison, status distribution



Monitoring

Longitudinal weekly and daily records per patient



Impact summary

Paired baseline-vs-latest analysis with dispersion



Formula tracking

Which version and batch each patient received



Export

Multi-sheet dataset, analysis-ready, plus formatted report

Statistically honest by design: paired comparisons only, dispersion reported with every mean, and study limitations printed on the same page as the results.

Live pilot, real clinical use

Not a prototype — running in an oncology centre today

45

patients enrolled

118

products manufactured
& dispatched

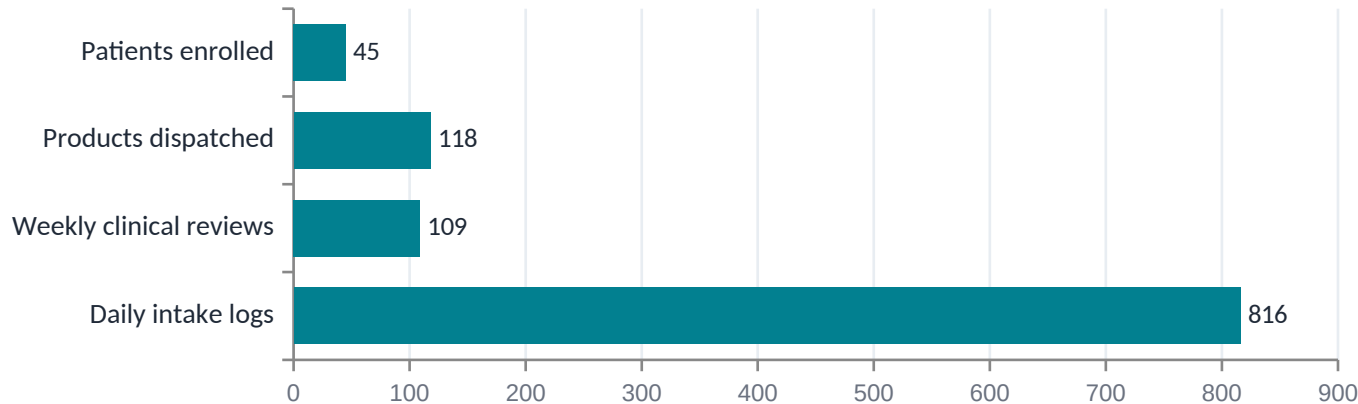
925

monitoring records
captured

7

weeks maximum
follow-up to date

Data captured during the pilot



What the pilot proves

- Clinicians will use it in a live clinic
- The manufacturing chain works end to end
- Longitudinal data accumulates reliably
- The engine handles real case complexity

How outcomes are reported

Generated live from the platform — and deliberately conservative

Paired baseline vs latest

Measure	Reported as
● Weight	mean change ± standard error
● Albumin	mean change ± standard error
● CRP	inverted — falling is improvement
● MUAC	mean change ± standard error
● Grip strength	mean change ± standard error

Headline metric: proportion of patients who held or gained weight — because in advanced cancer, maintenance is the therapeutic goal.

Stated limitations

Printed on the same page as the results — never in a footnote:

- Single-arm pilot — no control group, so change cannot be attributed to the supplement alone
- Patients receive concurrent oncological treatment, which independently affects every measure
- Only patients with both a baseline and follow-up value count for each metric — n varies by row
- Albumin and CRP are acute-phase reactants and move with infection and treatment cycles

A tool that hides its limitations should not be trusted with clinical decisions.

Why this is different

The gap it fills is between calculation and care

● **Generic nutrition calculators**

Compute a target and stop. No oncology logic, no drug interactions, no follow-through.

● **Hospital EMR modules**

Record what was ordered. They do not compute the prescription, manufacture it, or measure adherence.

● **Paper-based dietetics**

Clinically sound but unscalable, inconsistent between practitioners, and invisible to research.

● **GQUENCE**

Computes, safety-checks, manufactures to batch, dispatches, tracks consumption and flags deterioration — one governed loop.

What we are asking

1

Clinical review

Assess the engine against your own protocols and tell us where it is wrong.

2

Extended pilot

A larger cohort with structured follow-up, to move from engagement data to outcome evidence.

3

Site partnership

Multi-centre deployment — the platform already supports per-site cohort comparison.

Ready today

- Live in a production clinic
- Six governed user roles
- Full audit trail per patient
- Research export built in
- Configurable clinical rules